

Internship Report
Of
A Study on Enterprise Networking Infrastructure at Cisco Systems

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
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Section: 3CSE9
Batch:
2026-27

CANDIDATE'S DECLARATION

I, Yash Pandey, Roll Number 2410031674, do hereby declare that the internship report titled “A Study on Enterprise Networking Infrastructure at Cisco Systems” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Yash Pandey

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Section: 3CSE9

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout my internship at Cisco Systems. I am thankful for their aspiring guidance, invaluable constructive criticism, and friendly advice during this work. I am sincerely grateful to my mentors and the technical team at Cisco for sharing their truthful and illuminating views on a number of issues related to networking and this project.

I would also like to thank the faculty of the School of Computer Science and Engineering, IILM University, for their continuous support and encouragement throughout the duration of this internship.

I express my gratitude to my parents for their blessings.

Signature

Student Name: Yash Pandey
Roll No.: 2410031674

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3. INTERNSHIP COMPLETION CERTIFICATE

[This page is reserved for the original Internship Completion Certificate issued by Cisco Systems. Please scan and insert the certificate image here before final submission / printing.]



4. PROJECT DESCRIPTION

4.1 Introduction

This report presents an account of the internship undertaken at Cisco Systems, a global leader in networking hardware, software, and telecommunications equipment. The internship focused on enterprise networking concepts, including switching, routing, network security fundamentals, and the design and simulation of scalable network topologies using Cisco's industry-standard tools.

During the course of the internship, exposure was gained to real-world enterprise network architecture, structured cabling, VLAN segmentation, dynamic routing protocols, and basic network security practices such as access control lists and port security. The internship provided practical, hands-on experience that complemented the theoretical knowledge gained during the B.Tech Computer Science and Engineering curriculum.

The internship period allowed for direct engagement with networking professionals, participation in configuration exercises on Cisco IOS-based devices, and involvement in troubleshooting simulated network faults, thereby strengthening both technical and analytical skills relevant to the field of computer networking.

4.2 Organization Profile

Cisco Systems, Inc. is an American multinational technology conglomerate headquartered in San Jose, California. It is widely recognized as one of the world's leading developers and manufacturers of networking hardware, software, telecommunications equipment, and other high-technology products and services. Cisco's core product lines include routers, switches, wireless access points, security appliances, and collaboration tools that form the backbone of enterprise, service provider, and data center networks worldwide.

The organization is also known for its certification ecosystem, including CCNA (Cisco Certified Network Associate) and CCNP (Cisco Certified Network Professional), which are considered industry benchmarks for networking proficiency. Cisco's culture emphasizes innovation, collaboration, and continuous learning, and the internship

program is structured to give students practical exposure to enterprise-grade networking technologies alongside mentorship from experienced engineers.

4.3 Problem Statement

Organizations today rely on robust, secure, and scalable network infrastructure to support business-critical applications, communication, and data exchange. A poorly designed network can lead to bottlenecks, security vulnerabilities, downtime, and inefficient use of resources. The problem addressed during this internship was to design, simulate, and validate a scalable enterprise network topology that supports multiple departments through VLAN segmentation, ensures efficient inter-VLAN routing, and incorporates basic security measures, while maintaining high availability and ease of future expansion.

4.4 Project Objectives

- To understand the fundamentals of enterprise networking including the OSI and TCP/IP models.
- To design a multi-department LAN topology using Cisco switches and routers.
- To implement VLAN segmentation for logical separation of departmental traffic.
- To configure inter-VLAN routing using a router-on-a-stick and Layer 3 switching approach.
- To implement basic network security measures such as port security and access control lists (ACLs).
- To configure and compare dynamic routing protocols such as RIP, OSPF, and EIGRP.
- To simulate, test, and validate the designed network using Cisco Packet Tracer.
- To document the network design, configuration steps, and outcomes in a structured report.

4.5 Scope of the Project

The scope of the project is limited to designing and simulating an enterprise-style network for a hypothetical multi-department organization comprising departments such

as Administration, Human Resources, Engineering, and Sales. The project covers Layer 2 switching concepts (VLANs, trunking, spanning tree), Layer 3 routing (static and dynamic routing protocols), and fundamental network security configurations. The implementation and testing were carried out in a simulated environment using Cisco Packet Tracer rather than physical production hardware, which allowed safe experimentation without impacting live enterprise systems.

The project does not extend to wide area network (WAN) service provider configurations, advanced data center technologies, or cloud-native networking, which were outside the defined scope of the internship duration.

4.6 Technologies and Tools Used

The following technologies, tools, and protocols were used during the course of the internship:

S. No.	Category	Tools / Technologies
1	Simulation Software	Cisco Packet Tracer
2	Networking Devices	Cisco Catalyst Switches, Cisco ISR Routers
3	Operating System	Cisco IOS (Internetwork Operating System)
4	Switching Concepts	VLAN, Trunking (802.1Q), STP, Port Security
5	Routing Protocols	Static Routing, RIP v2, OSPF, EIGRP
6	Security	Access Control Lists (ACLs), Port Security, SSH
7	Addressing	IPv4 Addressing and Subnetting (VLSM)
8	Documentation	MS Word, MS Visio / Draw.io

Table 4.1: Technologies and Tools Used During Internship

4.7 System Architecture

The proposed network architecture follows a hierarchical three-layer design comprising the Core, Distribution, and Access layers, which is a widely adopted best practice in enterprise networking recommended by Cisco. The Access layer connects end-user devices within each department; the Distribution layer aggregates traffic from multiple access switches and performs inter-VLAN routing and policy enforcement; and the Core layer provides high-speed backbone connectivity between distribution blocks.

- Core Layer: A central router/multilayer switch providing high-speed backbone connectivity.
- Distribution Layer: Layer 3 switches performing inter-VLAN routing, ACL enforcement, and redundancy.
- Access Layer: Layer 2 switches connecting end devices (PCs, laptops, printers) within each VLAN.

Fig 4.1: Hierarchical Network Architecture (Core – Distribution – Access)

Each department was assigned a separate VLAN and IP subnet using Variable Length Subnet Masking (VLSM) to optimize address utilization. Table 4.2 summarizes the VLAN and IP addressing scheme used in the design.

VLAN ID	Department	Subnet	Gateway
10	Administration	192.168.10.0/26	192.168.10.1
20	Human Resources	192.168.20.0/26	192.168.20.1
30	Engineering	192.168.30.0/25	192.168.30.1
40	Sales	192.168.40.0/26	192.168.40.1

Table 4.2: VLAN and IP Addressing Scheme

4.8 Methodology

The project was carried out in a structured, phased manner as outlined below:

- Requirement Analysis: Identifying departmental needs, expected number of hosts, and growth considerations.
- Network Design: Creating the logical topology diagram with device placement, VLAN scheme, and IP addressing plan.
- Device Configuration: Configuring hostname, VLANs, trunk and access ports, and interface IP addresses on switches and routers.
- Routing Configuration: Implementing inter-VLAN routing and testing static and dynamic routing protocols (RIP, OSPF, EIGRP) for connectivity between subnets.
- Security Implementation: Applying port security to restrict unauthorized devices and configuring standard/extended ACLs to control traffic between departments.

- Testing and Validation: Using ping, traceroute, and the Packet Tracer simulation mode to verify end-to-end connectivity and correct traffic filtering.
- Documentation: Recording configurations, topology diagrams, and test results for reporting purposes.

This phased methodology ensured that each layer of the network was validated before progressing to the next, minimizing configuration errors and simplifying troubleshooting.

4.9 Expected Outcomes

- A fully functional simulated enterprise network with logically segmented departments via VLANs.
- Successful inter-VLAN communication through properly configured Layer 3 routing.
- Improved network security through port security and access control list implementation.
- A comparative understanding of the behaviour and convergence time of RIP, OSPF, and EIGRP.
- Hands-on proficiency in Cisco IOS command-line configuration relevant to CCNA-level competencies.
- A well-documented network design that can be scaled for additional departments or sites in the future.

4.10 Certificates of Completion and Other Communication / Mail Proof

[This section is reserved for attaching the internship completion certificate, offer/joining letter, and any relevant email communication exchanged with the organization during the internship. Please insert scanned copies or screenshots here before final submission.]

CISCO SYSTEMS INDIA PVT. LTD.

To: Yash Pandey (Roll No.: 2410031674)
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

Date: May 10, 2026

Subject: Offer Letter for Network Engineering Internship

Dear Yash,

We are pleased to offer you an internship position as a **Network Engineering Intern** at **Cisco Systems India Pvt. Ltd.** This offer is contingent upon your continued academic enrollment at IILM University.

1. Internship Overview

Position Title	Network Engineering Intern
Department / Team	Enterprise Networking Infrastructure Team
Supervisor / Mentor	Senior Network Architect, Cisco Systems
Duration & Location	6 Months (Batch 2026–2027) Cisco Campus / Remote Simulation Lab

2. Key Responsibilities

- Study enterprise-grade hierarchical network designs (Core, Distribution, and Access layers).
- Design, configure, and test multi-department LAN topologies using Cisco Packet Tracer and Cisco IOS.
- Implement Layer 2 features (VLAN segmentation, 802.1Q trunking, Spanning Tree, Port Security).
- Configure Layer 3 inter-VLAN routing and dynamic routing protocols (RIP v2, OSPF, EIGRP).
- Deploy basic network security policies, including Access Control Lists (ACLs) and SSH management.
- Prepare structured project documentations, topology reports, and performance evaluations.

3. Terms & Conditions

- Confidentiality:** You agree not to disclose proprietary network design or technical details outside Cisco Systems or IILM University.
- Code of Conduct:** You are expected to adhere strictly to Cisco's Global Code of Conduct and maintain professional standards.
- Intellectual Property:** All project artifacts, reports, and configurations created during the internship remain the property of Cisco Systems and IILM University.

Please sign and return a duplicate copy to confirm your acceptance.

For Cisco Systems India Pvt. Ltd.

Accepted & Agreed

Authorized Signatory
University Relations & Talent Acquisition

Yash Pandey
Candidate Signature & Date

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